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An Unsupervised Anomaly Detection Approach Using DBSCAN

Abstract

Financial fraud inflicts billions in losses annually and erodes trust in financial systems, making robust detection critical in the financial and insurance sectors. The extreme imbalance in real-world datasets, with fraudulent transactions constituting only ~1.21% of cases, challenges traditional supervised learning methods, as they often produce biased models with high false-negative rates. Preliminary experiments using XGBoost on the banksim1 dataset achieved 79% precision and 70% recall but suffered from excessive false positives, which incur significant operational costs in high-stakes environments. To address this, we propose an unsupervised clustering approach using the Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm, which excels at identifying rare fraudulent transactions as outliers without requiring scarce labeled data, directly tackling the data imbalance issue. The Kaggle credit card fraud dataset was avoided due to its PCA-transformed features, which obscure interpretable attributes essential for feature engineering. This project aims to develop a scalable anomaly detection system by leveraging DBSCAN to analyze banksim1's granular transaction features, effectively flagging suspicious activities with reduced false positives.

1. Introduction

1.1. Problem Statement: The Pervasive Challenge of Financial Fraud and the Critical Need

for Effective Detection

Financial fraud poses a significant and escalating global challenge, inflicting substantial financial losses upon institutions and individuals, estimated at billions annually [9], while simultaneously eroding public trust in financial systems [16]. The rapid digitization of financial transactions, encompassing everything from online banking to mobile payments, has intensified the urgency for sophisticated, real-time detection systems. These systems must possess the agility to adapt to the dynamic and continuously evolving patterns of fraudulent activities. Traditional, static rule-based systems often prove inadequate in this landscape, as they struggle to keep pace with the sheer volume and complexity of modern transactions, frequently failing to identify novel fraud schemes [8].

A fundamental technical hurdle in developing effective AI-based fraud detection models is the extreme rarity of fraudulent transactions compared to legitimate ones. This leads to severely imbalanced datasets, where the minority class (fraud) may represent as little as 1.21% of the total data. This imbalance presents a profound challenge for machine learning algorithms, often resulting in models that are biased towards the majority class. Such bias can manifest as high false-negative rates, meaning a significant number of actual fraudulent transactions are missed, directly translating to unrecovered financial losses and reputational damage for financial institutions. The practical consequence of missed fraud underscores the critical imperative for more effective and robust detection methodologies.

1.2. Motivation: The Limitations of Traditional Methods, Particularly with Highly Imbalanced Fraud Data

The inherent shortcomings of traditional fraud detection methods, which are typically rigid and demand extensive manual oversight, have become increasingly apparent. These systems are often unable to scale with the burgeoning volume of digital transactions and struggle to detect emerging fraud tactics. Consequently, there has been a significant shift towards leveraging advanced data analytics and Artificial Intelligence (AI) to enhance corporate governance and internal control processes.

Despite the advancements in supervised machine learning, their application in fraud detection is significantly hampered by the data imbalance problem. While ensemble methods such as XGBoost are recognized for their ability to mitigate data imbalance and enhance model robustness, often achieving high AUC scores , their practical utility can be limited in scenarios of extreme imbalance. Preliminary experiments conducted as part of this project with XGBoost on the banksim1 dataset, for instance, yielded 79% precision and 70% recall. However, these results were accompanied by a high rate of false positives, which incur significant operational costs due to manual reviews [15]. This outcome demonstrates that while XGBoost is generally powerful [30], its performance for this specific, highly skewed dataset, where false positives carry a high operational cost [15], is not ideal. The practical burden of investigating numerous false alarms can outweigh the benefits of detecting a small number of true positives, leading to inefficiencies and increased operational costs [9].

Furthermore, the practical application of supervised learning and deep learning models is constrained by the scarcity and high cost of obtaining accurately labeled fraud data. Manual labeling by experts is a resource-intensive and time-consuming process, particularly for large and dynamic datasets. This economic and logistical constraint provides a compelling practical justification for exploring alternative approaches that do not rely on pre-labeled data, such as unsupervised learning. The field of financial fraud detection is continuously evolving towards more sophisticated optimization techniques and improved handling of data imbalance and noisy data, indicating an ongoing need for novel solutions beyond the current mainstream.

1.3. Project Objectives: Clearly State the Research Objectives

This research project aims to address the challenges by pursuing the following key objectives:

- To investigate and implement unsupervised anomaly detection techniques, with a primary focus on the DBSCAN algorithm, for the purpose of financial fraud prevention.
- To develop a robust and scalable anomaly detection system capable of effectively identifying rare fraudulent transactions within highly imbalanced datasets, such as banksim1. The emphasis is on creating a system that can discern subtle deviations from normal behavior, which are characteristic of fraud, rather than relying on predefined fraud labels.
- To rigorously evaluate the performance of the developed unsupervised system against established baselines, particularly the preliminary XGBoost results, utilizing relevant metrics such as precision, recall, and F1-score. A crucial aspect of this evaluation will be demonstrating improved fraud detection capabilities, specifically with a significant reduction in false positives, directly addressing the limitations observed in supervised methods.
- To analyze the efficacy of DBSCAN in discerning complex normal transaction patterns and accurately flagging outliers as potential fraud. This analysis will contribute to a deeper understanding of DBSCAN's applicability and performance in real-world financial scenarios, thereby helping to bridge a recognized research gap in the application of unsupervised learning for complex fraud patterns.

1.4. Project Contributions: Outline the Expected Contributions

This project is expected to deliver several significant contributions to the field of financial fraud detection:

- A novel application and thorough evaluation of DBSCAN for financial transaction fraud detection. This work specifically addresses the challenges posed by extreme data imbalance, offering a practical solution where traditional methods falter.
- The demonstration of a practical, unsupervised anomaly detection system that can effectively identify fraudulent activities without reliance on scarce and costly labeled data. This enhances the deployability and scalability of fraud detection solutions in real-world financial environments.

- The provision of a comprehensive comparative analysis between supervised (XGBoost) and unsupervised (DBSCAN) approaches within the context of highly imbalanced fraud data. This analysis will highlight the distinct advantages of density-based clustering for outlier detection, particularly in minimizing false positives.
- A valuable contribution to the broader academic understanding of unsupervised learning's potential in critical domains like financial security. By rigorously exploring and validating DBSCAN in this context, the project directly addresses the identified research gap where unsupervised learning, especially clustering algorithms, remains underrepresented in recent literature concerning complex fraud patterns. This work will provide empirical evidence supporting the utility of such methods.

2. Literature Review and State-of-the-Art

2.1. Overview of Fraud Detection Techniques

The landscape of fraud detection has undergone a significant evolution, transitioning from rudimentary rule-based systems to sophisticated approaches leveraging advanced data analytics and Artificial Intelligence (AI). Early fraud detection mechanisms primarily relied on predefined rules, which, while simple to implement, proved rigid and incapable of adapting to the continuously evolving tactics employed by fraudsters. The increasing volume and complexity of digital transactions necessitated a shift towards more adaptive and intelligent systems capable of identifying subtle, hidden patterns that human-centric or manual systems might miss.

Machine learning (ML) has emerged as a powerful paradigm in this domain, broadly categorized into supervised, unsupervised, and semi-supervised learning. Supervised learning techniques, such as Logistic Regression, Support Vector Machines (SVM), and ensemble methods like XGBoost, are widely employed due to their high predictive power when ample labeled data is available. These methods learn to classify transactions as fraudulent or legitimate based on historical examples. Conversely, unsupervised learning operates without the need for labeled data, focusing on discovering inherent structures or anomalies within the data. This approach is particularly relevant for anomaly detection, where the goal is to model "normal" behavior and identify significant deviations from it. Semi-supervised methods combine elements of both, leveraging a small amount of labeled data alongside a larger pool of unlabeled data. The industry's recognition of the limitations of manual systems has driven the necessity for AI-driven solutions, making the exploration of advanced machine learning techniques highly pertinent to current needs.

2.2. Challenges of Data Imbalance in Fraud Detection

Data imbalance constitutes a fundamental and pervasive challenge in the development and deployment of AI-driven fraud detection models. In most financial datasets, fraudulent transactions represent an extremely small fraction of all transactions, often less than 1.5%. This severe class imbalance leads to biased models that predominantly favor the majority class (legitimate transactions), resulting in compromised fraud detection capabilities and, critically, high false-negative rates. A high false-negative rate implies that a significant number of actual fraudulent activities are misclassified as legitimate, leading to substantial financial losses for institutions.

Various mitigation strategies have been explored within supervised learning to address this issue:

- **Oversampling techniques**, such as Synthetic Minority Oversampling Technique (SMOTE), aim to increase the representation of the minority class by generating synthetic samples. While these methods can significantly improve recall rates, excessive oversampling may introduce noise into the dataset, potentially leading to marginal decreases in precision.
- **Under sampling** the majority class is another approach to balance the dataset. However, this comes at the cost of reducing the overall data completeness, potentially discarding valuable information about legitimate transactions.
- **Cost-sensitive learning** approaches adjust misclassification penalties, assigning a higher

cost to false negatives (missing fraud) than to false positives (flagging legitimate transactions as fraud). This technique aims to achieve a more balanced precision-recall trade-off.

- **Ensemble methods**, such as boosting algorithms like XGBoost, are known for their ability to effectively mitigate data imbalance by aggregating multiple weak learners to enhance model robustness. These methods often achieve high Area Under the Curve (AUC) scores and demonstrate strong fraud detection performance across varying fraud prevalence rates.

Despite the general effectiveness of these mitigation strategies and the robustness of ensemble methods like XGBoost, the extreme imbalance inherent in real-world fraud datasets can still pose significant practical challenges. As observed in preliminary experiments for this project, even with XGBoost, a high volume of false positives can persist. This indicates that while these methods are theoretically powerful, their practical utility in *extremely* imbalanced, high-stakes environments, where the cost of false positives is substantial, remains limited. The continuous emphasis on data imbalance across recent literature as a "fundamental challenge" and an area requiring "more sophisticated optimization techniques" underscores that a truly robust and universally effective solution remains an active area of research, particularly for unlabeled and highly skewed data.

2.3. Supervised vs. Unsupervised Learning for Anomaly Detection

The choice between supervised and unsupervised learning paradigms is pivotal in anomaly detection, especially within the context of financial fraud. Each approach possesses distinct advantages and disadvantages that dictate its suitability for specific problem characteristics.

Supervised Learning methods, while excelling at classification tasks when ample labeled data is available, face inherent limitations in fraud detection. Their reliance on pre-labeled fraud instances makes them vulnerable to concept drift, where evolving fraud patterns render previously learned models obsolete. Furthermore, the scarcity, cost, and time-consuming nature of obtaining accurately labeled fraud data pose significant practical barriers to their widespread application and continuous adaptation. Deep learning models, despite their capacity to process vast amounts of data and capture intricate patterns, also share these limitations, alongside high computational requirements and a lack of transparency.

In contrast, **Unsupervised Learning** methods offer a compelling alternative for anomaly detection. These approaches do not require labeled data, instead focusing on identifying deviations from established "normal" patterns. This characteristic makes them inherently ideal for rare event detection, such as fraud, and highly adaptive to new or evolving fraud types that have not been seen before. The absence of a dependency on labeled data translates into significant practical advantages, including reduced manual workload for data annotation, enhanced real-time detection capabilities, and substantial cost savings by minimizing the number of cases requiring further manual inspection.

A critical research gap exists in the comprehensive application of unsupervised learning techniques, particularly clustering algorithms, to address the complexities of financial fraud patterns. Despite their clear potential, unsupervised methods remain underrepresented in recent literature, necessitating further research to identify the most effective approaches for increasingly complex and evolving fraud scenarios. Real-world case studies, such as the deployment of AI-driven unsupervised fraud detection in health insurance, have demonstrated tangible benefits, including a 25% reduction in fraudulent claims and significant cost savings, validating the practical efficacy of this paradigm. This evidence suggests that while supervised methods are well-researched, unsupervised learning offers a powerful, yet underexplored, avenue for more adaptive and scalable fraud detection solutions.

Table 1: Comparison of Fraud Detection Approaches (Supervised vs. Unsupervised)

Feature/Criterion	Supervised Learning (e.g., XGBoost)	Unsupervised Learning (e.g., DBSCAN)
Data Requirement	Labeled fraud and non-fraud data (scarce, costly)	Unlabeled data; learns from normal patterns
Data Imbalance	Major challenge; leads to biased models, high false negatives	Handles naturally by identifying outliers as noise
Fraud Definition	Learns from known fraud patterns; struggles with new types	Detects deviations from normal; adaptive to evolving fraud
False Positives	Can be high, especially with imbalanced data (Preliminary Results)	Aims to minimize false positives by focusing on confident outliers
Computational Needs	Can be high for deep learning	Varies; DBSCAN can be efficient for density-based clustering
Interpretability	Often a "black box" for complex models	Outliers are explicitly flagged as noise (DBSCAN) or deviations
Research Status	More commonly employed; well-researched	Underrepresented; significant research gap

2.4. DBSCAN and Clustering Algorithms in Fraud Detection

Clustering algorithms, a subset of unsupervised learning, are designed to group similar data points together, thereby uncovering inherent structures or patterns within unlabeled datasets. In the context of fraud detection, clustering can be instrumental in identifying dense regions of "normal" behavior and isolating sparse, suspicious regions that represent potential anomalies.

Among various clustering algorithms, DBSCAN (Density-Based Spatial Clustering of Applications with Noise) stands out as particularly suitable for anomaly detection in financial transactions.[3] Its core principle involves grouping together closely packed data points, termed "core samples," and marking as outliers those points that lie alone in low-density regions, which are explicitly treated as "noise". This characteristic is highly advantageous for fraud detection, as rare fraudulent transactions naturally manifest as such noise points, distinct from the dense clusters of legitimate activity.

DBSCAN offers several key advantages that make it a robust choice for this project:

- **No Pre-defined Number of Clusters:** Unlike algorithms such as K-means, DBSCAN does not require the number of clusters to be specified beforehand. This is a significant benefit in financial fraud detection, as the precise number of "normal" transaction patterns or behavioral groups is typically unknown and can be highly variable.
- **Discovery of Arbitrary Cluster Shapes:** DBSCAN is capable of identifying clusters of arbitrary shapes, a crucial feature for handling the often "messy" and irregularly distributed real-world financial transaction data. This contrasts with methods like K-means, which assume spherical clusters and can struggle with complex, non-convex data distributions. This ability to adapt to the true underlying structure of the data makes DBSCAN more robust.
- **Natural Outlier Identification:** The algorithm's inherent mechanism for identifying "noise points" as outliers aligns perfectly with the definition of rare fraudulent transactions. By explicitly labeling these sparse points, DBSCAN provides a direct and intuitive flag for potential anomalies.

Academic literature supports the application of DBSCAN for segmenting transaction landscapes and isolating suspicious regions. This establishes a strong precedent for its use in modeling normal activity and effectively identifying deviations, thereby providing a solid foundation for the methodology proposed in this project. The explicit mention of DBSCAN in the context of advanced fraud detection frameworks further reinforces its relevance and potential impact.

3. Project Progress and Proposed Methodology

This section details the progress made in the project to date, outlining the dataset selection, preliminary experiments with a supervised baseline, and the rationale behind the shift to the

proposed unsupervised clustering methodology.

3.1. Dataset Selection and Characteristics

For this project, the banksim1 dataset was selected as the primary data source after a careful evaluation of available datasets for financial fraud detection. The decision prioritized the dataset's structure and suitability for feature engineering and interpretability, critical for developing a robust anomaly detection system. The Kaggle credit card fraud dataset, a common benchmark, was explicitly ruled out because its features are PCA-transformed, obscuring original, interpretable attributes such as transaction amount, time, or merchant type. This transformation severely limits opportunities for meaningful feature engineering, such as deriving behavioral signals like average spending, deviations from historical patterns, or transaction timing irregularities, which are essential for understanding and interpreting fraudulent behaviors in real-world financial analysis.

In contrast, the banksim1 dataset provides granular, interpretable features vital for a transparent and effective anomaly detection system. It contains approximately 594,643 transactions spanning 180 simulated days, with only 1.21% (~7,200 transactions) labeled as fraudulent, closely mirroring the extreme class imbalance observed in real-world financial datasets. This imbalance, where legitimate transactions dominate, is central to the project's motivation, as it reflects the challenge that unsupervised anomaly detection aims to address. The dataset is split into 80% for training (~475,714 transactions) and 20% for testing (~118,929 transactions) to ensure sufficient data for model development and robust evaluation. The structure and key features of the banksim1 dataset are detailed in Table 2.

Table 2: Banksim1 Dataset Features and Description

Feature Name	Description	Relevance to Fraud Detection
Step	Represents the simulated day within the dataset, spanning 180 steps over six months.	Captures temporal patterns, useful for identifying time-based anomalies or trends in fraudulent activity.
Customer	Unique identifier for individual customers involved in transactions.	Enables tracking customer-specific spending habits and deviations from their typical behavior.
ZipCodeOrigin	Indicates the origin postcode of each transaction.	Provides geographical context, useful for detecting unusual transaction locations or patterns.

Feature Name	Description	Relevance to Fraud Detection
Merchant	Unique identifier for merchants involved in transactions.	Allows for analysis of merchant-specific fraud patterns or unusual merchant activity.
ZipMerchant	Provides the postcode associated with each merchant.	Offers additional geographical context related to the merchant's location.
Age	Categorizes customer age into bands (e.g., '0' for ≤ 18, '1' for 19–25, 'U' for Unknown).	Provides demographic information that might correlate with certain transaction types or fraud profiles.
Gender	Categorizes customers as 'E' (Enterprise), 'F' (Female), 'M' (Male), or 'U' (Unknown).	Offers demographic insights, potentially revealing gender-specific fraud patterns.
Category	Captures the purchase category, indicating the type of transaction (e.g., travel, groceries).	Essential for identifying unusual spending categories for a customer or merchant.
Amount	Represents the monetary value of transactions.	A critical feature for detecting unusually large or small transactions, or deviations from typical spending amounts.
Fraud (or Class)	The binary target variable, where 1 indicates a fraudulent transaction and 0 indicates a non-fraudulent transaction.	The label used for initial baseline evaluation and for assessing the performance of the unsupervised model against known fraud.

The rich, interpretable features of banksim1 enable sophisticated feature engineering, such as creating temporal markers (e.g., hour, day of week, weekend indicators) and aggregated behavioral signals (e.g., average spending, deviation from historical patterns). These engineered features are expected to expose latent patterns indicative of fraudulent behaviors, enhancing the detection capabilities of the proposed DBSCAN-based system.

3.2. Initial Explorations with Supervised Learning

As an initial phase of the project, preliminary experiments were conducted using a supervised learning approach, specifically XGBoost, on the banksim1 dataset. XGBoost was chosen due to

its reputation as a robust ensemble method known for its strong performance on tabular data and its capabilities in mitigating data imbalance. The objective of this preliminary exploration was to establish a baseline performance and to empirically understand the challenges posed by the highly imbalanced nature of the dataset.

The experiments yielded a precision of 79% and a recall of 70%. While these numerical metrics might appear reasonable in isolation, a deeper analysis revealed a critical practical limitation: the high frequency of false positives. Despite XGBoost's ability to achieve decent classification scores, the sheer volume of legitimate transactions incorrectly flagged as fraudulent posed a significant operational burden. In a real-world financial setting, each false positive necessitates manual review by fraud analysts, leading to increased operational costs, delays in legitimate transactions, and potential customer dissatisfaction. This outcome underscored that while supervised models can learn from existing fraud labels, their practical utility for this specific problem instance, characterized by extreme imbalance and a low tolerance for false alarms, is significantly hampered. The observed false positive rate indicated that the model was not "ideal for this task," despite its general strengths.

3.3. Rationale for Unsupervised Clustering Approach

The limitations observed in preliminary supervised learning experiments with XGBoost on the banksim1 dataset, particularly the high rate of false positives (79% precision, 70% recall), underscored the challenges of applying supervised methods to extremely imbalanced datasets (1.21% fraud). This prompted a pivot to an unsupervised clustering approach, with the Density-Based Spatial Clustering of Applications with Noise (DBSCAN) selected as the primary algorithm for its alignment with the project’s goals.

DBSCAN was chosen for its key strengths in addressing financial fraud detection challenges:

- **No Labeled Data Requirement**: DBSCAN does not rely on pre-labeled fraud data, which is scarce and costly in banksim1, where fraud constitutes only 1.21% of transactions. This enhances scalability and adaptability to evolving fraud patterns.
- **Inherent Outlier Detection**: DBSCAN identifies outliers as “noise” points, perfectly suited for detecting rare fraudulent transactions that deviate from normal behavior, unlike supervised methods that struggle with minority class classification.
- **Handling Irregular Cluster Shapes**: Unlike K-means, which assumes spherical clusters and a fixed number of clusters, DBSCAN discovers clusters of arbitrary shapes, making it robust for the complex, non-linear patterns in financial transaction data.
- **Reduced False Positives**: By modeling dense regions of legitimate transactions, DBSCAN aims to minimize false positives, addressing the operational burden observed in XGBoost’s high false positive rate.

Other unsupervised methods, such as Gaussian Mixture Models (GMM) and HDBSCAN, were considered. GMM models data as a mixture of Gaussian distributions but may struggle with non-Gaussian transaction patterns, while HDBSCAN, a hierarchical extension of DBSCAN, offers flexibility in cluster density but lacks DBSCAN's explicit noise detection, which is critical for flagging rare fraud cases. K-means was ruled out due to its assumption of spherical clusters, unsuitable for banksim1's messy data. DBSCAN's explicit outlier identification and flexibility with cluster shapes make it the most suitable choice, though HDBSCAN will be explored as a secondary baseline in Section 3.5 to validate this decision.

3.4. System Architecture and Components

The proposed anomaly detection system is designed as a comprehensive, modular framework tailored for financial fraud prevention. It comprises several key components, each responsible for a distinct phase of the anomaly detection pipeline:

1. Data Preparation Module:
    - This module collects the raw banksim1 dataset and performs essential data cleaning and preprocessing steps, including handling missing values, one-hot encoding categorical features (e.g., Category, Gender), and standardizing numerical features (e.g., Amount) to ensure compatibility with clustering algorithms.
    - A critical aspect is feature engineering, leveraging banksim1's interpretable features detailed in Table 2 (Section 3.1). These include Amount (transaction value), Step (used to derive temporal markers like hour, day of week, and weekend indicators), Category (purchase type, e.g., travel, groceries), Customer (unique ID), Merchant (unique ID), ZipCodeOrigin, ZipMerchant, Age (bands), and Gender (categories). Engineered features include aggregated metrics such as per-customer average transaction amount, deviation from historical spending patterns, and transaction frequency per category. These features expose latent behavioral patterns indicative of fraud, enhancing the model's ability to discern anomalies.
2. Clustering Model (DBSCAN Implementation):
    - This core component implements the DBSCAN algorithm, trained on the preprocessed and engineered banksim1 transaction data to group normal transactions into dense clusters.
    - DBSCAN's parameters, *eps* (maximum distance between two samples for one to be considered in the neighborhood of another) and *min_samples* (minimum number of samples in a neighborhood for a point to be a core point), will be systematically tuned using techniques like k-distance graphs to optimize clustering performance. This tuning ensures effective delineation of normal

transaction patterns and accurate identification of outliers.

- Transactions not belonging to any dense cluster, marked as "noise" by DBSCAN, are flagged as potential fraud cases due to their deviation from established norms.

3. Anomaly Detection Engine:
    - This module processes DBSCAN's output to evaluate new, incoming transactions in real-time or near real-time. Each transaction is assessed against the trained DBSCAN model to determine if it falls within a dense cluster of normal activity or is classified as a noise point.
    - No additional thresholding is applied beyond DBSCAN's inherent noise detection mechanism, which relies on the tuned *eps* and *min_samples* parameters to identify outliers. Transactions flagged as noise are marked as suspicious and trigger alerts.
    - The engine provides a user-friendly interface, such as a dashboard, displaying key details of flagged transactions (e.g., Amount, Category, Customer ID, Step-derived temporal markers, and deviation metrics like spending anomalies). This interface enables fraud analysts to prioritize and review alerts efficiently, facilitating rapid investigation and response. For example, an alert might highlight a transaction with an unusually high Amount in an atypical Category for a specific Customer, prompting further scrutiny.
4. Evaluation Framework:
    - This module rigorously assesses the anomaly detection system's performance using standard metrics: precision (proportion of flagged transactions that are fraudulent), recall (proportion of actual fraudulent transactions identified), and F1-score (harmonic mean of precision and recall). These metrics are critical for evaluating effectiveness in imbalanced datasets like banksim1.
    - Results from the DBSCAN-based system will be compared against the preliminary XGBoost results (79% precision, 70% recall) to highlight improvements, particularly in reducing false positives, which address the operational burden observed in supervised methods.

3.5. Experimental Setup and Evaluation Framework

The experimental setup for the DBSCAN-based anomaly detection system is designed to ensure robust testing and evaluation, primarily using the banksim1 dataset (~594,643 transactions, ~7,200 fraudulent, 1.21% fraud rate) with its interpretable features (Step, Customer, Amount, Category, Age, Gender; see Table 2, Section 3.1). To enhance result confidence, a secondary experiment will be conducted using the Kaggle credit card fraud dataset (~284,807 transactions, ~492 fraudulent, 0.17% fraud rate), despite its PCA-transformed features limiting feature engineering.

Data preprocessing for banksim1 includes one-hot encoding for categorical variables (e.g., Category, Gender) and standardization for numerical features (e.g., Amount). Feature engineering will derive temporal features from Step (e.g., hour of day, day of week, weekend indicators) and aggregated metrics (e.g., per-customer average transaction amount, deviation from historical spending, transaction frequency per category) to capture behavioral patterns indicative of fraud. The Kaggle dataset will undergo similar preprocessing, though its PCA-transformed features restrict behavioral feature derivation. Both datasets will be split into 80% training (~475,714 for banksim1, ~227,846 for Kaggle) and 20% testing (~118,929 for banksim1, ~56,961 for Kaggle) to ensure sufficient data for model development and evaluation.

The DBSCAN algorithm will be implemented using standard machine learning libraries, with *eps* (maximum distance for neighborhood inclusion) and *min_samples* (minimum points for a core point) tuned systematically via k-distance graphs or grid search to optimize outlier detection. To validate DBSCAN's effectiveness, an Isolation Forest model, a robust unsupervised anomaly detection method that uses tree-based partitioning to isolate anomalies, will be implemented as a baseline for both datasets. Isolation Forest is chosen for its efficiency in high-dimensional data and its ability to handle imbalanced datasets without requiring labeled data, providing a strong comparison to DBSCAN's density-based approach.

Performance will be evaluated using standard metrics for imbalanced classification and anomaly detection:

- **Precision**: Proportion of flagged transactions that are actually fraudulent, critical to minimize operational costs from false positives.
- **Recall**: Proportion of actual fraudulent transactions identified, vital to reduce financial losses from missed fraud.
- **F1-score**: Harmonic mean of precision and recall, balancing accuracy in imbalanced datasets.

These metrics will be calculated using the Fraud label in both datasets as ground truth. The evaluation will compare:

- **XGBoost Baseline**: Preliminary results (79% precision, 70% recall on banksim1) showed high false positives, creating operational inefficiencies. DBSCAN aims to reduce false positives by focusing on confident outliers, potentially improving practical utility even if precision or recall is lower. For example, a slight decrease in recall (e.g., 65%) could be acceptable if false positives are significantly reduced (e.g., by 50%), as this lowers manual review workload, a key concern in real-world fraud detection.

- **Isolation Forest Baseline**: This comparison will assess whether DBSCAN's density-based clustering outperforms Isolation Forest's tree-based anomaly detection in capturing banksim1's complex transaction patterns and handling Kaggle's PCA-transformed data. Isolation Forest may excel in scalability but could struggle with non-linear patterns compared to DBSCAN's ability to model arbitrary cluster shapes.

If DBSCAN does not outperform XGBoost in precision or recall, it is not inherently "bad" but reflects trade-offs. A lower recall with significantly fewer false positives could be preferable in high-stakes environments where manual investigation costs are substantial, aligning with the project's goal of operational efficiency. Conversely, if DBSCAN yields higher false positives, it may indicate suboptimal parameter tuning or feature engineering, necessitating further refinement. The Isolation Forest comparison will clarify whether DBSCAN's density-based approach is superior for fraud detection, while the Kaggle dataset experiment will test robustness across different data structures, acknowledging PCA limitations. Results will highlight DBSCAN's ability to handle extreme data imbalance (1.21% fraud in banksim1, 0.17% in Kaggle) and adaptability to evolving fraud patterns, contributing to a scalable, label-free solution

4. Preliminary Results and Discussion

4.1. Analysis of Initial XGBoost Results

The preliminary experiments with XGBoost on the banksim1 dataset yielded a precision of 79% and a recall of 70%. While these figures indicate a reasonable ability to identify fraudulent transactions, a deeper examination reveals a significant practical limitation, particularly for real-world deployment in financial institutions. The primary concern arising from these results is the substantial number of false positives, which necessitate costly manual investigations [15].

In a domain like financial fraud detection, the cost associated with false positives is considerable, requiring human analyst review and increasing operational expenses [9]. Each instance where a legitimate transaction is incorrectly flagged as fraudulent necessitates manual investigation by a human analyst. This process is not only resource-intensive and time-consuming but can also lead to delays in processing legitimate transactions, causing customer dissatisfaction and potentially eroding trust in the financial institution [16]. Given that legitimate transactions vastly outnumber fraudulent ones (1.21% fraud rate), even a relatively high precision percentage can translate into a large absolute number of false positives. For example, if a model has 79% precision on a dataset where 98.79% of transactions are legitimate, a significant volume of non-fraudulent transactions will still be flagged, creating an overwhelming workload for human review.

This operational burden highlights that while XGBoost is a powerful algorithm generally effective at mitigating data imbalance [30], its performance in this specific, extremely skewed context, where false positive tolerance is low [6], is not optimal.The model's tendency to generate frequent false alarms underscores the inherent challenges of applying supervised classification directly to highly imbalanced anomaly detection problems, motivating the exploration of alternative paradigms.

4.2. Expected Outcomes from DBSCAN

Based on the inherent properties of DBSCAN and its established applications in anomaly detection, the proposed unsupervised approach is anticipated to yield significant improvements, particularly in addressing the limitations observed with supervised methods.

The primary expected outcome is a substantial reduction in false positives. DBSCAN's core mechanism of identifying outliers as "noise" points, rather than classifying them based on learned patterns of a rare minority class, is inherently better suited for distinguishing true anomalies from the vast majority of normal transactions. By focusing on the density of data points, it is expected to form robust clusters of legitimate behavior, thereby more accurately identifying transactions that genuinely deviate from the norm. This targeted approach is designed to minimize the misclassification of legitimate transactions, leading to fewer

unnecessary alerts and a more efficient allocation of investigative resources. This aligns with the practical benefits observed in other unsupervised fraud detection applications, which have demonstrated a reduction in manual review workload and improved operational efficiency.

Furthermore, the DBSCAN approach is expected to demonstrate improved adaptability to evolving fraud patterns. Unlike supervised models that learn from historical fraud labels and may struggle with novel fraud types (concept drift), DBSCAN models normal behavior. Any new, unseen fraudulent activity that deviates significantly from established normal patterns will naturally be flagged as an outlier, even if it doesn't resemble previously known fraud. This inherent flexibility provides a more robust and future-proof solution for dynamic financial environments.

Finally, the project anticipates demonstrating the **practical viability of** unsupervised learning for financial fraud detection without reliance on labeled data. This will not only contribute to filling the identified research gap in the application of clustering algorithms for complex fraud patterns but also offer a scalable and cost-effective solution for financial institutions. The ability to detect fraud effectively without the continuous, expensive process of manual data labeling represents a significant practical advantage, leading to potentially substantial cost savings and faster claim settlements, as evidenced in related industry case studies.

5. Reflection on Project Plan and Future Work

5.1. Progress Against Agreed Plan

Significant progress has been made on the project to date, aligning with the mid-point expectations of the agreed project plan. The initial phases involved a comprehensive literature review, which successfully established a robust theoretical foundation for the project by contextualizing the challenges of financial fraud and the state-of-the-art in detection techniques. This review thoroughly explored the limitations of traditional and supervised machine learning methods, particularly concerning data imbalance, and identified a critical research gap in the application of unsupervised learning for this domain.

Furthermore, practical work has been undertaken, including the meticulous selection and initial characterization of the banksim1 dataset. Preliminary experiments with XGBoost were successfully conducted to establish a performance baseline and to empirically validate the challenges associated with supervised learning on highly imbalanced data. This foundational work has directly informed the subsequent strategic decisions regarding the project's methodological direction.

5.2. Justification for Adjustments

A notable adjustment to the initially planned objectives for the project involves a decisive pivot from a primary focus on supervised learning to a dedicated unsupervised clustering approach, specifically utilizing DBSCAN. This change was not arbitrary but a direct consequence of the empirical findings from the preliminary XGBoost experiments and a deeper analysis of the problem's inherent characteristics.

The initial XGBoost model, despite its general robustness for imbalanced data, yielded a high rate of false positives on the banksim1 dataset. This practical limitation highlighted the operational inefficiencies and costs associated with such a system in a real-world financial context. The decision to shift was driven by the understanding that for extremely imbalanced datasets where fraud is truly an outlier (1.21% prevalence), supervised classification, even with mitigation strategies, struggles to achieve the desired balance between precision and recall without generating an unmanageable number of false alarms.

Concurrently, the literature review solidified the understanding of the "critical research gap" in unsupervised learning for fraud detection and underscored the inherent advantages of density-based clustering algorithms like DBSCAN. DBSCAN's ability to identify outliers as noise without requiring labeled data, its adaptability to arbitrary cluster shapes, and its potential for minimizing false positives directly address the shortcomings observed in the supervised baseline. This adjustment ensures the project remains focused on developing a solution that is not only theoretically sound but also practically viable and

impactful for financial fraud prevention. The choice of the banksim1 dataset over the PCA-transformed Kaggle dataset was also a justified adjustment, prioritizing interpretable features necessary for effective feature engineering and actionable insights.

5.3. Remaining Work and Next Steps

The remaining phase of the project will focus on the full implementation, optimization, and comprehensive evaluation of the proposed unsupervised anomaly detection system. The immediate next steps include:

- **Detailed Data Preparation and Feature Engineering:** Completing the robust data preprocessing pipeline for the banksim1 dataset, with a strong emphasis on generating rich, behavioral features as outlined in Section 3.1. This will involve creating temporal features, aggregating customer and merchant specific transaction patterns, and exploring other domain-relevant transformations.
- **DBSCAN Model Implementation and Parameter Tuning:** Implementing the DBSCAN algorithm and systematically tuning its eps and min_samples parameters. This will involve iterative experimentation and validation to identify the optimal configuration that effectively identifies fraudulent transactions as noise while minimizing false positives.
- **Development of Anomaly Detection Engine:** Building the module responsible for processing new transactions, applying the trained DBSCAN model, and generating real-time or near real-time alerts for suspicious activities.
- **Comprehensive Evaluation and Comparative Analysis:** Rigorously evaluating the performance of the DBSCAN-based system using precision, recall, and F1-score. This will include a detailed comparative analysis against the established XGBoost baseline, highlighting the improvements achieved by the unsupervised approach in handling data imbalance and reducing false positives.
- **Visualization and Interpretability:** Developing visualizations to illustrate the clustering results and the separation of anomalies, enhancing the interpretability of the model's findings. This will aid in understanding the characteristics of flagged transactions.
- **Documentation and Report Finalization:** Documenting all methodologies, experimental setups, results, and analyses in detail, culminating in the final project report.

6. Conclusion

This mid-term report details the significant progress made in developing an advanced anomaly detection system for financial fraud prevention. The pervasive challenge of financial fraud, exacerbated by the extreme data imbalance inherent in transaction datasets, was thoroughly established as the core problem. Preliminary explorations with a supervised learning approach, XGBoost, while yielding a 79% precision and 70% recall, critically highlighted the issue of frequent false positives, demonstrating its practical limitations for real-world deployment where operational efficiency is paramount.

In response to these challenges, a strategic pivot to an unsupervised clustering methodology, specifically utilizing DBSCAN, has been justified. This approach leverages DBSCAN's unique ability to model normal transaction patterns and identify rare outliers as noise, effectively bypassing the need for scarce labeled data and inherently aiming to minimize false positives. This methodological shift is not merely a technical choice but a problem-driven one, directly addressing the practical shortcomings of traditional methods and aligning with a recognized research gap in the academic literature concerning unsupervised learning for complex fraud patterns.

The project has successfully completed its foundational phases, including a comprehensive literature review, detailed dataset selection (banksim1) with a clear rationale for its choice over PCA-transformed alternatives, and the establishment of a supervised baseline. The remaining work is clearly defined, focusing on the full implementation, rigorous evaluation, and optimization of the DBSCAN-based system. The anticipated outcomes include a substantial reduction in false positives, improved adaptability to evolving fraud tactics, and the demonstration of a scalable, practical solution for financial institutions that does not rely on costly labeled data. This project is poised to contribute significantly to both the academic understanding and practical application of unsupervised learning in the critical domain of financial security.

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